Surrogate-model comparison — heteroscedastic mixed-variable test problems

Separate GPs / Categorical GP / Standard LVGP (MATLAB) / H-LVGP (MATLAB)

Legend. Metrics on a ground-truth test set (200-pt grid per level for 1-D; 1024 Sobol points per level for Rastrigin 6-D): MAE and RRMSE (surface accuracy), 95% Gneiting–Raftery interval score IS (accuracy + calibration, $\alpha = 0.05$), and 95% coverage (target 0.95; higher is *not* better). **Bold on green** = best model in the comparison group (winners are recomputed within each version's model set). Budgets n are TOTAL design points; designs are SLHD with n_{rep} replicates per point. Single seed.

Test problems — true functions and noise

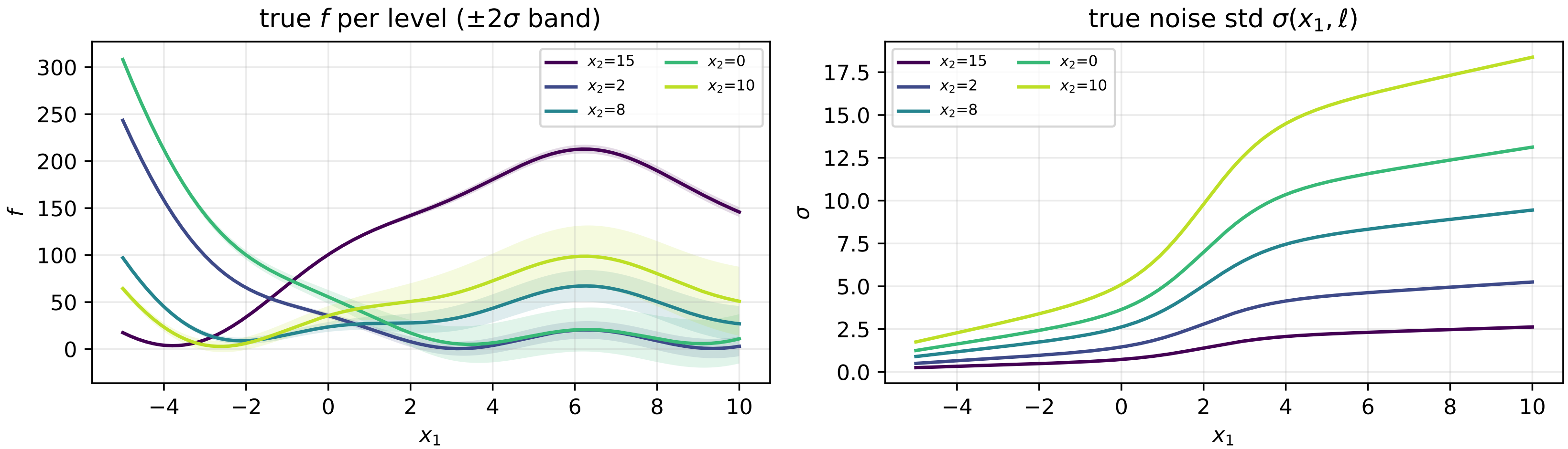

Figure 1: Branin (v2 noise): true response per level with $\pm 2\sigma$ noise band (left) and the region-contrast noise std — low-noise region, sigmoid step at $x_1 \approx 2$, high-noise region — with level multipliers $\{0.5, 1.0, 1.8, 2.5, 3.5\}$ (right).

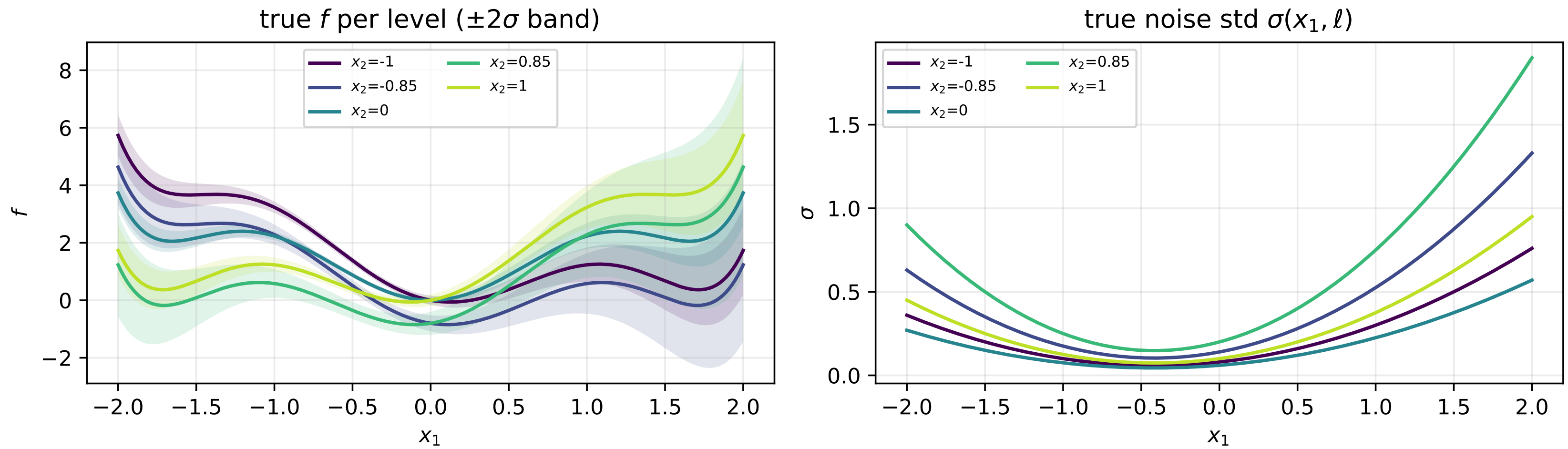

Figure 2: Six-hump camel (v4 levels $\{-1.0, -0.85, 0.0, 0.85, 1.0\}$, v4.3 polynomial noise $(0.04 + 0.05x_1 + 0.06x_1^2)m(\ell)$): two pairs $\{1, 2\}$ and $\{4, 5\}$ (within-pair curve RMS 0.82) plus a middle bridge level 3 (1.16–1.27 from everything; cross-pair 1.97–2.32). The 30-seed camel tables were run under the earlier v4.0 noise $0.05e^{(0.4x_1)^2}m(\ell)$.

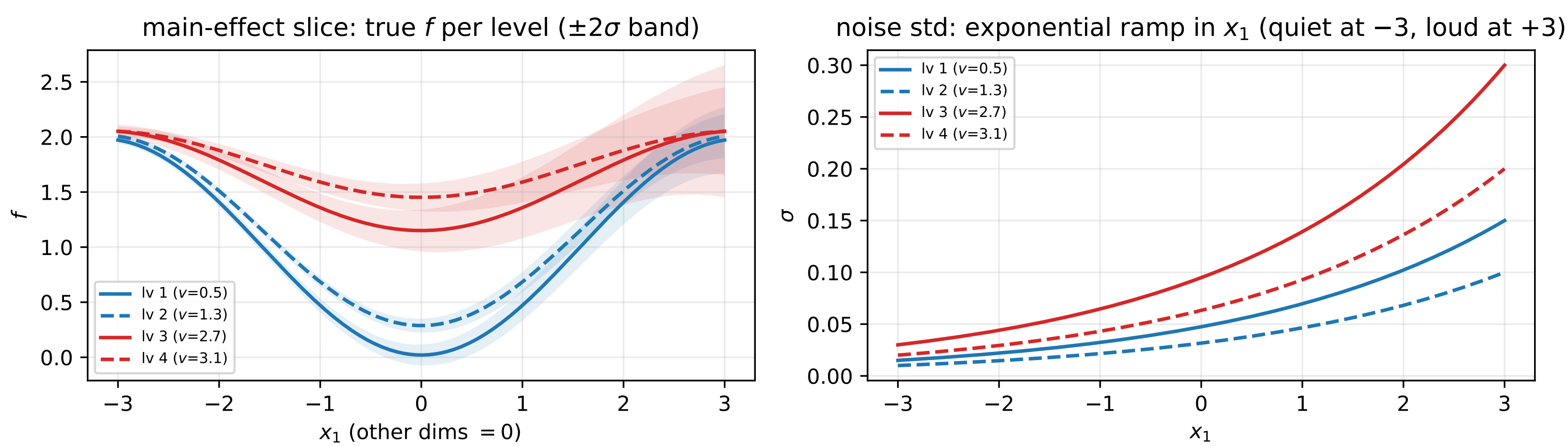

Figure 3: Griewank 6-D (v3.1, domain $[-3, 3]^5$): main-effect slice along x_1; levels form two pairs $\{0.5, 1.3\}$ and $\{2.7, 3.1\}$ (solid/dashed = pair members) separated by per-level offsets $b(\ell) \in \{0, 0.15, 0.6, 0.75\}$, with cluster-split noise multipliers $\{1.5, 1.0\}$ vs $\{3.0, 2.0\}$ and the exponential-ramp noise $0.01 \cdot 10^{(x_1+3)/6}m(\ell)$ (quiet at $x_1 = -3$, up to $\sigma = 0.30$ at $+3$).

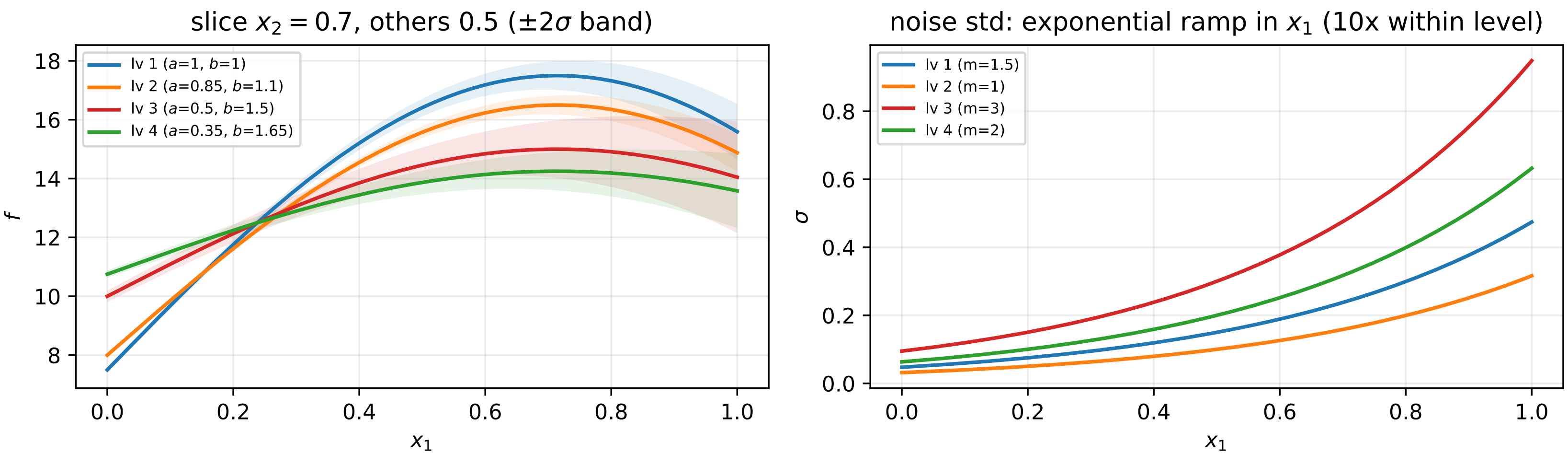

Figure 4: Friedman 5-D + categorical (TP-8, *candidate — single-seed pilot only*): levels scale the interaction amplitude $a(\ell) \in \{1.0, 0.85, 0.5, 0.35\}$ and the x_4 slope $b(\ell) \in \{1.0, 1.1, 1.5, 1.65\}$ — shape-only pairs $\{1,2\}/\{3,4\}$ (within-pair curve RMS 0.65/0.66, cross-pair 1.65–2.88), no constant offsets. Noise: exponential ramp $0.1 \cdot 10^{x_1 - 0.5} m(\ell)$, $m \in \{1.5, 1.0, 3.0, 2.0\}$.

Version 1 — all four models

Table 1: Mean-over-levels metrics, 1-D problems: problem × budget × model (seed 1). Bold on green = best model in each (problem, budget, n_{rep}) block per metric (MAE/RRMSE/IS: lower; coverage: closest to 0.95).

			$n_{\text{rep}} = 10$			
Problem	n	Model	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
Branin	25	Separate GPs	11.24	0.4550	125.4	0.7340
		Categorical GP	4.757	0.1821	41.85	**0.9340**
		Standard LVGP	3.932	0.1763	63.11	0.3710
		H-LVGP	**2.554**	**0.0835**	**28.91**	1.000
	40	Separate GPs	4.295	0.2002	41.91	**0.9360**
		Categorical GP	1.879	0.0652	**14.08**	1.000
		Standard LVGP	2.706	0.1087	55.61	0.4250
		H-LVGP	**1.350**	**0.0546**	30.60	1.000
Six-hump camel	25	Separate GPs	0.3120	0.3954	2.439	0.9340
		Categorical GP	0.2092	0.3095	**1.801**	**0.9560**
		Standard LVGP	**0.1844**	**0.2902**	4.264	0.4320
		H-LVGP	0.2331	0.3552	2.178	0.9350
	40	Separate GPs	0.1851	0.2875	1.942	**0.9410**
		Categorical GP	0.1261	0.1839	**1.068**	0.9590
		Standard LVGP	0.1229	**0.1685**	3.179	0.3070
		H-LVGP	**0.1197**	0.1690	1.441	1.000

Table 2: Mean-over-levels metrics, high-dimensional problems: problem × budget × model (seed 1). Bold on green = best model in each (problem, budget, n_{rep}) block per metric (MAE/RRMSE/IS: lower; coverage: closest to 0.95).

			$n_{\text{rep}} = 10$			
Problem	n	Model	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
Griewank 6-D	80	Separate GPs	0.0717	1.038	0.7414	0.8638
		Categorical GP	**0.0589**	**0.9533**	**0.6838**	**0.9014**
		Standard LVGP	0.1190	1.634	1.600	0.7559
		H-LVGP	0.0784	1.068	0.9081	0.8337
	120	Separate GPs	0.0628	0.8884	0.7160	0.8789
		Categorical GP	**0.0494**	**0.7336**	**0.5382**	**0.9246**
		Standard LVGP	0.0825	1.142	0.6957	0.8345
		H-LVGP	0.1012	1.347	1.082	0.8613
	160	Separate GPs	0.0547	0.8347	0.6527	0.8450
		Categorical GP	**0.0466**	**0.6850**	**0.5736**	0.8904
		Standard LVGP	0.0701	0.9459	0.5782	0.9006
		H-LVGP	0.1046	1.664	0.9121	**0.9207**
Friedman 5-D (pilot)	40	Separate GPs	2.653	0.6639	23.95	0.7153
		Categorical GP	**0.8860**	**0.2401**	**8.108**	**0.8669**
		Standard LVGP	1.343	0.3568	19.92	0.5764
		H-LVGP	1.305	0.3481	16.66	0.6704
	60	Separate GPs	1.875	0.4837	17.33	0.7542
		Categorical GP	**0.4697**	**0.1310**	**3.241**	**0.9719**
		Standard LVGP	0.5107	0.1410	3.857	0.8379
		H-LVGP	0.5373	0.1523	3.379	0.9048
	80	Separate GPs	1.265	0.3418	7.985	0.9243
		Categorical GP	**0.3186**	0.0998	2.107	**0.9688**
		Standard LVGP	0.3815	0.1041	2.251	0.8691
		H-LVGP	0.3329	**0.0969**	**2.001**	0.9888

Conclusions (preliminary — seed 1, $n_{\text{rep}} = 10$)

Status. Problem versions: Branin v2 (region-contrast noise), camel v4 (5 levels: two pairs + a middle bridge), Griewank 6-D v3.1 (level-shifted pairs, tamed domain $[-3,3]^5$, exponential-ramp noise $0.01 \cdot 10^{(x_1+3)/6} m(\ell)$ — quiet at $x_1 = -3$, up to $\sigma = 0.30$ at $+3$). All three problems now carry a 30-seed fixed-design replication (Table 3); Griewank uses the budget ladder 20/30/40 pts/level ($n \in \{80, 120, 160\}$). *Camel version caveat:* the 30-seed camel tables use the v4.0 noise $0.05e^{(0.4x_1)^2} m(\ell)$; the single-seed camel tables use the current v4.3 polynomial $(0.04 + 0.05x_1 + 0.06x_1^2)m(\ell)$ — a design exploration that left the model ranking essentially unchanged (noise treatment buys calibration, not mean accuracy).

1. **Cross-level information sharing dominates at these budgets.** The Separate GPs trail on accuracy in every configuration (roughly 2–4× the best MAE on the 1-D problems); their honest, wide intervals score well only on pure-calibration metrics.

2. **Branin v2, with 30-seed error bars: the H-LVGP's $n = 25$ accuracy win is robust; its $n = 40$ lead was a seed-1 artifact.** At $n = 25$ the H-LVGP dominates accuracy across noise realizations (MAE 2.52 ± 0.72 vs Categorical GP 4.53 ± 0.24) with coverage 1.00; the Categorical GP posts the study's best-calibrated result there (coverage 0.9455 ± 0.010). At $n = 40$ the means converge (1.59 ± 0.27 vs 1.72 ± 0.42, medians tied) and the Categorical GP clearly wins IS (14.9 vs 35.6).

3. **The camel arc: falsified hypothesis, synthesis, then tempered by replication.** The design sequence (v2.0 tight pairs → v2.1 widened → v3 hierarchy → v4 five levels) first suggested the latent needs ≥5 structured levels and ≥8 pts/level to pay (seed 1's v4 $n = 40$: H-LVGP 0.074 vs 0.101). **The 30-seed replication shows that was a favorable draw**: over noise realizations the Categorical GP leads camel accuracy at both budgets ($n = 40$: 0.114 ± 0.010 vs 0.143 ± 0.024) while the **H-LVGP is the best-calibrated model** (coverage 0.953 ± 0.029). Revised synthesis: at camel-scale level separation the latent buys *calibration*, not accuracy; Branin-scale separation ($n = 25$) is where it buys both.

4. **The standard LVGP's intervals are unusable everywhere** — coverage 0.22–0.78 against the 0.95 target (camel v3 $n = 32$: 0.290, NLL 3351), the worst miscalibration areas, and NLL up to thousands vs $O(1)$ for every other model. Its posterior mean is often competitive — the deficiency is purely the noise-blind uncertainty.

5. **Noise-surface recovery shows the polynomial aleatoric models' structural limits.** Branin v2's sigmoid step is outside both polynomial families (nMAE 0.25–0.33; the pooled $[x, z]$ polynomial's usual advantage vanishes), while on smooth noise the pooled version wins. When under-determined (36 coefficients vs 20–32 points) the pooled polynomial explodes out-of-sample (nMAE up to ≈1300); once over-determined it recovers the noise surface well.

6. **Griewank 6-D: offset differences favor the exchangeable kernel; shape differences favor the latent.** Isolation experiments identified the surface as the original bottleneck (budget ladder and noise rescale left RRMSE unchanged; the $[-3,3]^5$ domain fix made it learnable). The v2 redesign added TP-7-style per-level offsets $b(\ell)$ to separate the levels — and flipped every budget to the Categorical GP. Mechanism: the LVGP's single constant mean forces pure offsets through the latent distances, sacrificing cross-level shape-sharing. **The 30-seed budget ladder (v3.1 ramp noise, 20/30/40 pts/level) localizes the resolution threshold and exposes an H-LVGP pathology.** The Categorical GP crosses the mean-prediction ceiling already at 30/level (RRMSE 0.75 ± 0.02; 0.70 ± 0.05 at 40/level, MAE 0.0474 ± 0.003) and sweeps every budget; the Separate GPs follow (0.90/0.85); the standard LVGP stays at the ceiling until 40/level (1.05). **The H-LVGP's posterior mean degrades monotonically with n** — MAE 0.082 → 0.111 → 0.137, RRMSE 1.13 → 1.56 → 2.09

(median 2.15 at $n = 160$: the whole seed distribution, not outlier draws) — while its coverage stays acceptable (0.84–0.93) through inflated intervals. Every other model improves with data; the heteroscedastic machinery itself is implicated. This is the study's most actionable finding for the H-LVGP codebase and deserves a dedicated investigation (conditioning of the replicate-variance-in-correlation formulation at larger n is the prime suspect). No model reaches the 0.95 coverage target on griewank.

7. **Noise-awareness buys *reproducibility* of calibration, not just its level.** Across every replicated configuration the H-LVGP's coverage band is on-target *and* tight across noise realizations (seed-std 0.008–0.047), while the standard LVGP's calibration is erratic as well as biased: coverage seed-std 0.04–0.15, up to 18× larger (branin $n = 40$: ±0.145 vs ±0.008) — which side of 0.5 its coverage lands on depends on the noise draw. Its IS spread is likewise 1.5–3× the H-LVGP's on camel and griewank. Mechanism: replicate-variance-informed intervals adapt to each realization; a single MLE nugget re-absorbs whatever the draw looked like, making calibration a lottery. Scope: the Separate GPs' band is equally tight (±0.01–0.03) but off-target — the H-LVGP is the only model both centered and stable; and the stability does *not* extend to its posterior mean (next item).

8. **The LVGPs are 2–8× less stable run-to-run than the python GPs** — seed-to-seed std across every replicated configuration (Branin $n = 25$ MAE: H-LVGP ±0.72, standard ±1.06 vs CatGP ±0.24; Griewank $n = 160$: H-LVGP ±0.027 vs CatGP ±0.003): pure-MLE multistart variance vs prior-regularized fitting. Single-fit LVGP results carry real refit risk.

Practical recommendations.

regime	recommended model
many well-separated levels, strong structure (Branin-like), ≥5 pts/level	**H-LVGP** (best accuracy + calibrated intervals)
few (≤4) moderately structured levels, or ≤5 pts/level	**Categorical GP** (the latent needs ≥5 levels AND ≥8 pts/level to pay — camel v4, Branin)
intervals used downstream	never the standard LVGP alone
levels differing mainly by constant offsets	**Categorical GP** (the constant-mean LVGP pays for offsets in latent distance — Griewank v2/v3.1)
signal variation at/below the resolvable limit	fix the problem (scaling, domain, or function) first
high-n heteroscedastic fits with the current H-LVGP	audit first — its mean degrades with n on Griewank (conclusion 6)

Table 3: 30-seed replication, 1-D problems (fixed design, fresh noise replicates per seed): mean ± std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95).

Problem	n	Model	MAE	RRMSE	IS	Coverage
Branin	25	Separate GPs	10.91 ± 0.466	0.434 ± 0.022	113.7 ± 11.60	0.775 ± 0.028
		Categorical GP	4.529 ± 0.241	0.159 ± 0.011	38.87 ± 2.018	**0.946 ± 0.010**
		Standard LVGP	4.419 ± 1.063	0.212 ± 0.051	74.80 ± 30.69	0.442 ± 0.143
		H-LVGP	2.520 ± 0.723	**0.096 ± 0.037**	37.28 ± 20.11	0.996 ± 0.016
		H-LVGP (torch)	3.752 ± 0.611	0.136 ± 0.029	24.61 ± 2.495	0.960 ± 0.023
		H-LVGP (torch-mix)	**2.427 ± 0.510**	0.096 ± 0.030	**18.89 ± 0.894**	0.995 ± 0.014
	40	Separate GPs	4.454 ± 0.361	0.208 ± 0.017	42.88 ± 1.784	**0.931 ± 0.011**
		Categorical GP	1.585 ± 0.268	0.054 ± 0.012	14.86 ± 0.896	0.996 ± 0.010
		Standard LVGP	3.109 ± 0.705	0.122 ± 0.030	62.82 ± 20.09	0.410 ± 0.145
		H-LVGP	1.722 ± 0.425	0.064 ± 0.019	35.61 ± 10.83	0.998 ± 0.008
		H-LVGP (torch)	1.409 ± 0.195	0.048 ± 0.008	14.04 ± 1.000	0.976 ± 0.009
		H-LVGP (torch-mix)	**1.183 ± 0.191**	**0.040 ± 0.007**	**11.32 ± 0.756**	0.987 ± 0.011
Six-hump camel	25	Separate GPs	0.231 ± 0.009	0.319 ± 0.007	2.072 ± 0.061	**0.952 ± 0.008**
		Categorical GP	0.168 ± 0.009	0.262 ± 0.010	1.778 ± 0.136	0.923 ± 0.012
		Standard LVGP	**0.152 ± 0.018**	**0.242 ± 0.029**	3.405 ± 0.521	0.417 ± 0.097
		H-LVGP	0.178 ± 0.012	0.287 ± 0.016	2.153 ± 0.231	0.905 ± 0.017
		H-LVGP (torch)	0.231 ± 0.019	0.344 ± 0.022	2.211 ± 0.224	0.917 ± 0.022
		H-LVGP (torch-mix)	0.196 ± 0.017	0.283 ± 0.020	**1.582 ± 0.166**	0.942 ± 0.017
	40	Separate GPs	0.196 ± 0.008	0.295 ± 0.006	2.313 ± 0.111	0.888 ± 0.015
		Categorical GP	**0.114 ± 0.010**	**0.160 ± 0.015**	1.127 ± 0.164	0.918 ± 0.013
		Standard LVGP	0.145 ± 0.034	0.212 ± 0.066	3.240 ± 1.092	0.324 ± 0.106
		H-LVGP	0.143 ± 0.024	0.226 ± 0.042	1.147 ± 0.318	**0.953 ± 0.029**
		H-LVGP (torch)	0.174 ± 0.017	0.237 ± 0.019	1.449 ± 0.214	0.915 ± 0.022
		H-LVGP (torch-mix)	0.145 ± 0.019	0.197 ± 0.022	**1.107 ± 0.204**	0.928 ± 0.025

Table 4: 30-seed replication, high-dimensional problems (fixed design, fresh noise replicates per seed): mean ± std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95).

Problem	n	Model	MAE	RRMSE	IS	Coverage
Griewank 6-D	48	Separate GPs	0.076 ± 0.003	1.086 ± 0.036	0.833 ± 0.034	0.861 ± 0.014
		Categorical GP	**0.066 ± 0.003**	**0.939 ± 0.048**	**0.695 ± 0.027**	**0.893 ± 0.015**
		Standard LVGP	0.104 ± 0.009	1.376 ± 0.134	1.385 ± 0.193	0.716 ± 0.052
		H-LVGP	0.097 ± 0.007	1.242 ± 0.169	0.939 ± 0.070	0.845 ± 0.021
		H-LVGP (torch)	0.093 ± 0.004	1.145 ± 0.040	1.249 ± 0.079	0.707 ± 0.023
		H-LVGP (torch-mix)	0.082 ± 0.008	1.103 ± 0.119	0.869 ± 0.118	0.814 ± 0.038
	64	Separate GPs	0.066 ± 0.002	0.973 ± 0.025	0.773 ± 0.031	0.882 ± 0.010
		Categorical GP	**0.064 ± 0.003**	**0.896 ± 0.041**	**0.748 ± 0.022**	**0.912 ± 0.008**
		Standard LVGP	0.086 ± 0.012	1.210 ± 0.168	1.203 ± 0.273	0.736 ± 0.043
		H-LVGP	0.084 ± 0.026	1.141 ± 0.321	0.916 ± 0.226	0.876 ± 0.029
		H-LVGP (torch)	0.071 ± 0.003	1.002 ± 0.047	0.921 ± 0.035	0.828 ± 0.011
		H-LVGP (torch-mix)	0.068 ± 0.003	0.961 ± 0.044	0.818 ± 0.044	0.856 ± 0.016
	80	Separate GPs	0.071 ± 0.002	1.016 ± 0.040	0.730 ± 0.028	0.871 ± 0.010
		Categorical GP	**0.061 ± 0.003**	**0.986 ± 0.046**	**0.723 ± 0.024**	**0.887 ± 0.009**
		Standard LVGP	0.100 ± 0.010	1.359 ± 0.145	1.084 ± 0.228	0.820 ± 0.047
		H-LVGP	0.082 ± 0.015	1.133 ± 0.277	0.950 ± 0.089	0.836 ± 0.047
		H-LVGP (torch)	0.086 ± 0.004	1.116 ± 0.059	0.897 ± 0.045	0.825 ± 0.024
		H-LVGP (torch-mix)	0.068 ± 0.004	1.093 ± 0.057	0.771 ± 0.055	0.873 ± 0.023
	120	Separate GPs	0.064 ± 0.003	0.903 ± 0.036	0.690 ± 0.063	0.880 ± 0.017
		Categorical GP	**0.050 ± 0.001**	**0.749 ± 0.020**	0.539 ± 0.015	**0.922 ± 0.008**
		Standard LVGP	0.101 ± 0.015	1.382 ± 0.186	1.283 ± 0.269	0.773 ± 0.048
		H-LVGP	0.111 ± 0.033	1.564 ± 0.605	1.101 ± 0.182	0.856 ± 0.031
		H-LVGP (torch)	0.073 ± 0.002	0.996 ± 0.044	0.677 ± 0.030	0.863 ± 0.013
		H-LVGP (torch-mix)	0.054 ± 0.003	0.765 ± 0.040	**0.535 ± 0.027**	0.921 ± 0.015
	160	Separate GPs	0.056 ± 0.002	0.851 ± 0.040	0.683 ± 0.052	0.836 ± 0.018
		Categorical GP	**0.047 ± 0.003**	**0.704 ± 0.053**	0.598 ± 0.024	0.886 ± 0.011
		Standard LVGP	0.075 ± 0.008	1.048 ± 0.107	0.948 ± 0.391	0.811 ± 0.100
		H-LVGP	0.137 ± 0.027	2.091 ± 0.433	1.047 ± 0.132	**0.934 ± 0.020**
		H-LVGP (torch)	0.061 ± 0.002	0.812 ± 0.029	0.751 ± 0.044	0.822 ± 0.018
		H-LVGP (torch-mix)	0.050 ± 0.003	0.726 ± 0.051	**0.587 ± 0.031**	0.878 ± 0.013
Friedman 5-D	40	Separate GPs	2.656 ± 0.015	0.664 ± 0.004	23.97 ± 0.275	0.716 ± 0.009
		Categorical GP	0.883 ± 0.011	0.239 ± 0.003	7.819 ± 0.256	**0.873 ± 0.007**
		Standard LVGP	1.218 ± 0.227	0.327 ± 0.051	15.52 ± 4.699	0.683 ± 0.102
		H-LVGP	1.242 ± 0.183	0.332 ± 0.042	14.92 ± 3.072	0.712 ± 0.067
		H-LVGP (torch)	1.239 ± 0.175	0.329 ± 0.039	14.19 ± 2.913	0.727 ± 0.063
		H-LVGP (torch-mix)	**0.662 ± 0.026**	**0.177 ± 0.007**	**5.375 ± 0.374**	0.864 ± 0.013
		Separate GPs	1.858 ± 0.047	0.480 ± 0.011	16.44 ± 1.483	0.768 ± 0.022
		Categorical GP	0.483 ± 0.028	0.135 ± 0.008	3.286 ± 0.223	0.963 ± 0.007

Table 5: 30-seed replication, 1-D problems (fixed design, fresh noise replicates per seed): mean ± std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95). Without the Categorical GP (winners recomputed).

Problem	n	Model	MAE	RRMSE	IS	Coverage
Branin	25	Separate GPs	10.91 ± 0.466	0.434 ± 0.022	113.7 ± 11.60	0.775 ± 0.028
		Standard LVGP	4.419 ± 1.063	0.212 ± 0.051	74.80 ± 30.69	0.442 ± 0.143
		H-LVGP	2.520 ± 0.723	**0.096 ± 0.037**	37.28 ± 20.11	0.996 ± 0.016
		H-LVGP (torch)	3.752 ± 0.611	0.136 ± 0.029	24.61 ± 2.495	**0.960 ± 0.023**
		H-LVGP (torch-mix)	**2.427 ± 0.510**	0.096 ± 0.030	**18.89 ± 0.894**	0.995 ± 0.014
	40	Separate GPs	4.454 ± 0.361	0.208 ± 0.017	42.88 ± 1.784	**0.931 ± 0.011**
		Standard LVGP	3.109 ± 0.705	0.122 ± 0.030	62.82 ± 20.09	0.410 ± 0.145
		H-LVGP	1.722 ± 0.425	0.064 ± 0.019	35.61 ± 10.83	0.998 ± 0.008
		H-LVGP (torch)	1.409 ± 0.195	0.048 ± 0.008	14.04 ± 1.000	0.976 ± 0.009
		H-LVGP (torch-mix)	**1.183 ± 0.191**	**0.040 ± 0.007**	**11.32 ± 0.756**	0.987 ± 0.011
Six-hump camel	25	Separate GPs	0.231 ± 0.009	0.319 ± 0.007	2.072 ± 0.061	**0.952 ± 0.008**
		Standard LVGP	**0.152 ± 0.018**	**0.242 ± 0.029**	3.405 ± 0.521	0.417 ± 0.097
		H-LVGP	0.178 ± 0.012	0.287 ± 0.016	2.153 ± 0.231	0.905 ± 0.017
		H-LVGP (torch)	0.231 ± 0.019	0.344 ± 0.022	2.211 ± 0.224	0.917 ± 0.022
		H-LVGP (torch-mix)	0.196 ± 0.017	0.283 ± 0.020	**1.582 ± 0.166**	0.942 ± 0.017
	40	Separate GPs	0.196 ± 0.008	0.295 ± 0.006	2.313 ± 0.111	0.888 ± 0.015
		Standard LVGP	0.145 ± 0.034	0.212 ± 0.066	3.240 ± 1.092	0.324 ± 0.106
		H-LVGP	**0.143 ± 0.024**	0.226 ± 0.042	1.147 ± 0.318	**0.953 ± 0.029**
		H-LVGP (torch)	0.174 ± 0.017	0.237 ± 0.019	1.449 ± 0.214	0.915 ± 0.022
		H-LVGP (torch-mix)	0.145 ± 0.019	**0.197 ± 0.022**	**1.107 ± 0.204**	0.928 ± 0.025

Table 6: 30-seed replication, high-dimensional problems (fixed design, fresh noise replicates per seed): mean ± std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95). Without the Categorical GP (winners recomputed).

Problem	n	Model	MAE	RRMSE	IS	Coverage
		Separate GPs	**0.076 ± 0.003**	**1.086 ± 0.036**	**0.833 ± 0.034**	**0.861 ± 0.014**
		Standard LVGP	0.104 ± 0.009	1.376 ± 0.134	1.385 ± 0.193	0.716 ± 0.052
	48	H-LVGP	0.097 ± 0.007	1.242 ± 0.169	0.939 ± 0.070	0.845 ± 0.021
		H-LVGP (torch)	0.093 ± 0.004	1.145 ± 0.040	1.249 ± 0.079	0.707 ± 0.023
		H-LVGP (torch-mix)	0.082 ± 0.008	1.103 ± 0.119	0.869 ± 0.118	0.814 ± 0.038
		Separate GPs	**0.066 ± 0.002**	0.973 ± 0.025	**0.773 ± 0.031**	**0.882 ± 0.010**
		Standard LVGP	0.086 ± 0.012	1.210 ± 0.168	1.203 ± 0.273	0.736 ± 0.043
	64	H-LVGP	0.084 ± 0.026	1.141 ± 0.321	0.916 ± 0.226	0.876 ± 0.029
		H-LVGP (torch)	0.071 ± 0.003	1.002 ± 0.047	0.921 ± 0.035	0.828 ± 0.011
		H-LVGP (torch-mix)	0.068 ± 0.003	**0.961 ± 0.044**	0.818 ± 0.044	0.856 ± 0.016
		Separate GPs	0.071 ± 0.002	**1.016 ± 0.040**	**0.730 ± 0.028**	0.871 ± 0.010
Griewank 6-D		Standard LVGP	0.100 ± 0.010	1.359 ± 0.145	1.084 ± 0.228	0.820 ± 0.047
	80	H-LVGP	0.082 ± 0.015	1.133 ± 0.277	0.950 ± 0.089	0.836 ± 0.047
		H-LVGP (torch)	0.086 ± 0.004	1.116 ± 0.059	0.897 ± 0.045	0.825 ± 0.024
		H-LVGP (torch-mix)	**0.068 ± 0.004**	1.093 ± 0.057	0.771 ± 0.055	**0.873 ± 0.023**
		Separate GPs	0.064 ± 0.003	0.903 ± 0.036	0.690 ± 0.063	0.880 ± 0.017
		Standard LVGP	0.101 ± 0.015	1.382 ± 0.186	1.283 ± 0.269	0.773 ± 0.048
	120	H-LVGP	0.111 ± 0.033	1.564 ± 0.605	1.101 ± 0.182	0.856 ± 0.031
		H-LVGP (torch)	0.073 ± 0.002	0.996 ± 0.044	0.677 ± 0.030	0.863 ± 0.013
		H-LVGP (torch-mix)	**0.054 ± 0.003**	**0.765 ± 0.040**	**0.535 ± 0.027**	**0.921 ± 0.015**
		Separate GPs	0.056 ± 0.002	0.851 ± 0.040	0.683 ± 0.052	0.836 ± 0.018
		Standard LVGP	0.075 ± 0.008	1.048 ± 0.107	0.948 ± 0.391	0.811 ± 0.100
	160	H-LVGP	0.137 ± 0.027	2.091 ± 0.433	1.047 ± 0.132	**0.934 ± 0.020**
		H-LVGP (torch)	0.061 ± 0.002	0.812 ± 0.029	0.751 ± 0.044	0.822 ± 0.018
		H-LVGP (torch-mix)	**0.050 ± 0.003**	**0.726 ± 0.051**	**0.587 ± 0.031**	0.878 ± 0.013
		Separate GPs	2.656 ± 0.015	0.664 ± 0.004	23.97 ± 0.275	0.716 ± 0.009
		Standard LVGP	1.218 ± 0.227	0.327 ± 0.051	15.52 ± 4.699	0.683 ± 0.102
	40	H-LVGP	1.242 ± 0.183	0.332 ± 0.042	14.92 ± 3.072	0.712 ± 0.067
		H-LVGP (torch)	1.239 ± 0.175	0.329 ± 0.039	14.19 ± 2.913	0.727 ± 0.063
		H-LVGP (torch-mix)	**0.662 ± 0.026**	**0.177 ± 0.007**	**5.375 ± 0.374**	**0.864 ± 0.013**
		Separate GPs	1.858 ± 0.047	0.480 ± 0.011	16.44 ± 1.483	0.768 ± 0.022
Friedman 5-D		Standard LVGP	0.505 ± 0.083	0.138 ± 0.022	4.128 ± 0.637	0.799 ± 0.048
	60	H-LVGP	0.552 ± 0.131	0.154 ± 0.035	3.529 ± 1.010	0.910 ± 0.019
		H-LVGP (torch)	0.504 ± 0.028	0.143 ± 0.010	3.473 ± 0.358	0.897 ± 0.018
		H-LVGP (torch-mix)	**0.377 ± 0.026**	**0.105 ± 0.008**	**2.283 ± 0.265**	**0.946 ± 0.016**
		Separate GPs	1.274 ± 0.014	0.345 ± 0.004	8.091 ± 0.122	0.919 ± 0.004
		Standard LVGP	0.348 ± 0.048	0.100 ± 0.014	2.638 ± 0.594	0.820 ± 0.043
	80	H-LVGP	0.345 ± 0.021	0.101 ± 0.006	2.018 ± 0.123	0.973 ± 0.013

Table 7: Branin — per-level metrics, $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.95). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

		Separate GPs				**Categorical GP**				**Standard LVGP**				**H-LVGP**			
n	Level	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
25	$x_2 = 15$	5.902	0.1174	35.77	0.6200	4.435	0.0967	**20.37**	**0.9350**	6.136	0.1679	96.80	0.4100	**3.339**	**0.0917**	32.68	1.000
	$x_2 = 2$	14.17	0.4118	162.3	0.7950	4.714	0.1723	49.29	**0.9150**	**2.032**	**0.0488**	31.46	0.4300	2.327	0.0531	**19.96**	1.000
	$x_2 = 8$	7.824	0.8427	130.3	0.8600	3.366	0.3004	**29.88**	**1.000**	3.853	0.3730	41.36	0.2900	**1.691**	**0.1164**	30.41	1.000
	$x_2 = 0$	16.19	0.2789	124.3	0.6850	6.556	0.1302	72.62	0.8200	**3.392**	0.0622	79.81	0.3450	3.565	**0.0615**	**28.74**	**1.000**
	$x_2 = 10$	12.14	0.6243	174.6	0.7100	4.714	0.2111	37.07	**1.000**	4.247	0.2297	66.10	0.3800	**1.847**	**0.0949**	**32.76**	1.000
	Mean	11.24	0.4550	125.4	0.7340	4.757	0.1821	41.85	**0.9340**	3.932	0.1763	63.11	0.3710	**2.554**	**0.0835**	**28.91**	1.000
40	$x_2 = 15$	**0.5195**	**0.0128**	8.882	**1.000**	0.8563	0.0214	7.364	1.000	0.7171	0.0222	**6.774**	0.8400	1.016	0.0368	30.29	1.000
	$x_2 = 2$	6.850	0.2434	85.51	**0.9100**	1.591	0.0525	**10.31**	1.000	3.729	0.1288	92.07	0.1650	**0.7563**	**0.0280**	25.59	1.000
	$x_2 = 8$	4.018	0.3915	38.87	**0.9050**	**0.7749**	**0.0480**	**13.18**	1.000	2.240	0.1564	27.19	0.4800	0.8975	0.0521	27.75	1.000
	$x_2 = 0$	3.333	0.0713	40.92	**0.9000**	2.576	0.0453	**17.40**	1.000	2.888	0.0597	45.91	0.3400	**1.096**	**0.0212**	34.81	1.000
	$x_2 = 10$	6.753	0.2818	35.39	**0.9650**	3.599	0.1588	**22.14**	1.000	3.958	0.1764	106.1	0.3000	**2.984**	**0.1347**	34.58	1.000
	Mean	4.295	0.2002	41.91	**0.9360**	1.879	0.0652	**14.08**	1.000	2.706	0.1087	55.61	0.4250	**1.350**	**0.0546**	30.60	1.000

Table 8: Six-hump camel — per-level metrics, $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.95). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

		Separate GPs				**Categorical GP**				**Standard LVGP**				**H-LVGP**			
n	Level	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
25	x_2 = -1	0.1689	0.2501	2.355	**0.9500**	**0.1501**	**0.2320**	**2.095**	0.9450	0.1532	0.2434	3.163	0.5500	0.1713	0.2904	2.601	0.9250
	x_2 = -0.85	0.2249	0.3515	3.078	0.9600	**0.1495**	**0.2485**	**1.687**	0.9600	0.1706	0.2957	4.223	0.6050	0.2511	0.2988	1.745	**0.9550**
	$x_2 = 0$	0.1822	0.4036	1.563	0.9800	0.1426	0.3728	**1.343**	0.9600	**0.1337**	**0.3440**	2.544	0.5050	0.1679	0.4539	1.749	**0.9450**
	$x_2 = 0.85$	0.6129	0.6476	3.049	**0.9550**	0.3698	0.3904	**1.776**	0.9700	**0.2916**	**0.3504**	7.908	0.1200	0.3352	0.4118	2.640	0.8900
	$x_2 = 1$	0.3711	0.3243	2.151	0.8250	0.2338	0.3040	**2.104**	**0.9450**	**0.1729**	**0.2178**	3.481	0.3800	0.2401	0.3208	2.156	0.9600
	Mean	0.3120	0.3954	2.439	0.9340	0.2092	0.3095	**1.801**	**0.9560**	**0.1844**	**0.2902**	4.264	0.4320	0.2331	0.3552	2.178	0.9350
40	x_2 = -1	0.1797	0.3163	2.472	**0.9350**	0.1213	0.1503	**0.9393**	0.9700	**0.0948**	**0.1151**	2.447	0.3600	0.1187	0.1803	1.245	1.000
	x_2 = -0.85	0.2187	0.3297	2.297	**0.9550**	**0.1404**	**0.1948**	**1.282**	0.9700	0.2121	0.3042	5.321	0.2750	0.1516	0.2230	1.854	1.000
	$x_2 = 0$	0.1888	0.3379	2.033	0.8550	0.1261	0.2380	1.010	0.8950	**0.0768**	**0.1458**	1.851	0.4000	0.0960	0.1751	**0.9064**	**1.000**
	$x_2 = 0.85$	0.1662	0.2421	1.571	**0.9950**	0.1191	0.1736	**1.069**	0.9950	0.1337	0.1614	3.791	0.1900	**0.1159**	**0.1433**	1.920	1.000
	$x_2 = 1$	0.1719	0.2113	1.338	**0.9650**	0.1238	0.1630	**1.039**	0.9650	**0.0972**	**0.1160**	2.485	0.3100	0.1162	0.1233	1.278	1.000
	Mean	0.1851	0.2875	1.942	**0.9410**	0.1261	0.1839	**1.068**	0.9590	0.1229	**0.1685**	3.179	0.3070	**0.1197**	0.1690	1.441	1.000

Table 9: Griewank 6-D — per-level metrics, $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.95). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

		Separate GPs				**Categorical GP**				**Standard LVGP**				**H-LVGP**			
n	Level	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
	x6 = 0.5	0.1124	1.039	1.094	**0.8887**	**0.0812**	**0.7925**	**1.017**	0.8350	0.1689	1.487	2.812	0.6250	0.1195	0.9957	1.528	0.6904
	x6 = 1.3	0.0844	0.8664	0.8943	0.8369	**0.0712**	**0.7962**	**0.8518**	**0.8623**	0.1409	1.420	2.162	0.6787	0.0939	0.9630	1.303	0.7451
80	x6 = 2.7	**0.0423**	0.9214	0.5231	0.8672	0.0434	**0.8983**	**0.4172**	**0.9541**	0.0811	1.481	0.7551	0.8555	0.0542	1.033	0.4656	0.9316
	x6 = 3.1	0.0478	1.326	0.4536	0.8623	**0.0398**	1.326	0.4488	**0.9541**	0.0852	2.148	0.6717	0.8643	0.0461	**1.280**	**0.3360**	0.9678
	Mean	0.0717	1.038	0.7414	0.8638	**0.0589**	**0.9533**	**0.6838**	**0.9014**	0.1190	1.634	1.600	0.7559	0.0784	1.068	0.9081	0.8337
	x6 = 0.5	0.1149	1.039	1.405	0.8096	**0.0723**	**0.6761**	**0.8521**	**0.8525**	0.1157	0.9518	1.175	0.7100	0.1506	1.229	1.783	0.7100
	x6 = 1.3	0.0656	0.7280	0.6875	**0.9229**	**0.0574**	**0.6359**	**0.6661**	0.8936	0.0918	0.8605	0.8627	0.7793	0.1186	1.175	1.418	0.7754
120	x6 = 2.7	0.0354	0.7809	0.3830	**0.9580**	**0.0343**	**0.6441**	**0.3109**	0.9814	0.0588	1.029	0.3864	0.9150	0.0709	1.299	0.6118	0.9688
	x6 = 3.1	0.0355	1.005	0.3884	0.8252	**0.0336**	**0.9784**	**0.3238**	0.9707	0.0636	1.727	0.3583	**0.9336**	0.0649	1.686	0.5136	0.9912
	Mean	0.0628	0.8884	0.7160	0.8789	**0.0494**	**0.7336**	**0.5382**	**0.9246**	0.0825	1.142	0.6957	0.8345	0.1012	1.347	1.082	0.8613
	x6 = 0.5	0.0731	0.7128	**0.7801**	**0.9141**	**0.0673**	**0.6724**	0.9984	0.7832	0.1006	0.8151	0.8253	0.8291	0.1295	1.170	1.190	0.8584
	x6 = 1.3	0.0658	0.7193	0.8450	0.8428	**0.0573**	**0.6520**	0.7931	0.8398	0.0807	0.7678	**0.6693**	**0.9014**	0.1072	1.113	1.069	0.8711
160	x6 = 2.7	0.0496	0.9630	0.4770	0.8779	**0.0328**	**0.6225**	**0.2719**	**0.9551**	0.0532	0.9725	0.4788	0.9082	0.0892	1.670	0.6728	0.9600
	x6 = 3.1	0.0303	0.9437	0.5085	0.7451	**0.0291**	**0.7932**	**0.2310**	0.9834	0.0457	1.228	0.3392	**0.9639**	0.0925	2.702	0.7174	0.9932
	Mean	0.0547	0.8347	0.6527	0.8450	**0.0466**	**0.6850**	**0.5736**	0.8904	0.0701	0.9459	0.5782	0.9006	0.1046	1.664	0.9121	**0.9207**

Table 10: Friedman 5-D (pilot) — per-level metrics, $n_{\mathrm{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.95). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

		Separate GPs				**Categorical GP**				**Standard LVGP**				**H-LVGP**			
n	Level	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
40	level 1	2.883	0.7168	35.53	0.5400	**1.633**	**0.4117**	**19.32**	**0.5771**	1.919	0.5056	37.19	0.4258	1.928	0.5072	34.30	0.4678
	level 2	3.648	0.9445	25.21	0.7588	**0.7032**	**0.2419**	**6.049**	**0.9346**	1.458	0.4125	19.22	0.5918	1.420	0.4047	16.31	0.6504
	level 3	2.100	0.5276	12.06	**0.9658**	**0.5622**	**0.1443**	**3.628**	0.9893	1.061	0.2778	14.27	0.5918	1.007	0.2648	9.959	0.7246
	level 4	1.980	0.4667	23.02	0.5967	**0.6449**	**0.1623**	**3.434**	**0.9668**	0.9341	0.2312	8.987	0.6963	0.8655	0.2157	6.081	0.8389
	Mean	2.653	0.6639	23.95	0.7153	**0.8860**	**0.2401**	**8.108**	**0.8669**	1.343	0.3568	19.92	0.5764	1.305	0.3481	16.66	0.6704
60	level 1	2.159	0.5576	14.19	0.8428	**0.5879**	**0.1749**	**4.084**	**0.9297**	0.6205	0.1800	5.883	0.7695	0.6499	0.1920	4.671	0.8506
	level 2	2.355	0.6243	23.10	0.6875	**0.5185**	**0.1520**	**3.228**	**0.9619**	0.5207	0.1553	4.299	0.8213	0.5457	0.1697	3.986	0.8486
	level 3	1.977	0.4900	25.71	0.5654	**0.3548**	**0.0920**	2.954	0.9990	0.4212	0.1098	**2.364**	0.8955	0.4548	0.1246	2.402	**0.9551**
	level 4	1.009	0.2630	6.315	0.9209	**0.4175**	**0.1051**	2.698	0.9971	0.4802	0.1189	2.882	0.8652	0.4990	0.1227	**2.457**	**0.9648**
	Mean	1.875	0.4837	17.33	0.7542	**0.4697**	**0.1310**	**3.241**	**0.9719**	0.5107	0.1410	3.857	0.8379	0.5373	0.1523	3.379	0.9048
80	level 1	1.456	0.4145	8.637	0.9531	**0.3405**	0.1243	2.613	**0.9473**	0.4100	0.1139	2.381	0.8740	0.3555	**0.1131**	**2.024**	0.9795
	level 2	1.656	0.4538	10.23	0.8770	**0.2954**	0.1054	2.161	**0.9551**	0.3519	**0.1006**	2.357	0.8340	0.3254	0.1023	**1.758**	0.9756
	level 3	1.157	0.3010	8.100	0.8789	0.3043	0.0813	1.903	1.000	0.3291	0.0890	**1.835**	**0.9023**	**0.2892**	**0.0786**	1.947	1.000
	level 4	0.7897	0.1980	4.970	0.9883	**0.3341**	**0.0882**	**1.750**	**0.9727**	0.4351	0.1130	2.430	0.8662	0.3615	0.0937	2.275	1.000
	Mean	1.265	0.3418	7.985	0.9243	**0.3186**	0.0998	2.107	**0.9688**	0.3815	0.1041	2.251	0.8691	0.3329	**0.0969**	**2.001**	0.9888

Version 2 — without the Categorical GP

Table 11: Mean-over-levels metrics, 1-D problems: problem × budget × model (without the Categorical GP) (seed 1). Bold on green = best model in each (problem, budget, n_{rep}) block per metric (MAE/RRMSE/IS: lower; coverage: closest to 0.95).

			$n_{\text{rep}} = 10$			
Problem	n	Model	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
Branin	25	Separate GPs	11.24	0.4550	125.4	0.7340
		Standard LVGP	3.932	0.1763	63.11	0.3710
		H-LVGP	**2.554**	**0.0835**	**28.91**	**1.000**
	40	Separate GPs	4.295	0.2002	41.91	**0.9360**
		Standard LVGP	2.706	0.1087	55.61	0.4250
		H-LVGP	**1.350**	**0.0546**	**30.60**	1.000
Six-hump camel	25	Separate GPs	0.3120	0.3954	2.439	0.9340
		Standard LVGP	**0.1844**	**0.2902**	4.264	0.4320
		H-LVGP	0.2331	0.3552	**2.178**	**0.9350**
	40	Separate GPs	0.1851	0.2875	1.942	**0.9410**
		Standard LVGP	0.1229	**0.1685**	3.179	0.3070
		H-LVGP	**0.1197**	0.1690	**1.441**	1.000

Table 12: Mean-over-levels metrics, high-dimensional problems: problem × budget × model (without the Categorical GP) (seed 1). Bold on green = best model in each (problem, budget, n_{rep}) block per metric (MAE/RRMSE/IS: lower; coverage: closest to 0.95).

			$n_{\text{rep}} = 10$			
Problem	n	Model	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
Griewank 6-D	80	Separate GPs	**0.0717**	**1.038**	**0.7414**	**0.8638**
		Standard LVGP	0.1190	1.634	1.600	0.7559
		H-LVGP	0.0784	1.068	0.9081	0.8337
	120	Separate GPs	**0.0628**	**0.8884**	0.7160	**0.8789**
		Standard LVGP	0.0825	1.142	**0.6957**	0.8345
		H-LVGP	0.1012	1.347	1.082	0.8613
	160	Separate GPs	**0.0547**	**0.8347**	0.6527	0.8450
		Standard LVGP	0.0701	0.9459	**0.5782**	0.9006
		H-LVGP	0.1046	1.664	0.9121	**0.9207**
Friedman 5-D (pilot)	40	Separate GPs	2.653	0.6639	23.95	**0.7153**
		Standard LVGP	1.343	0.3568	19.92	0.5764
		H-LVGP	**1.305**	**0.3481**	**16.66**	0.6704
	60	Separate GPs	1.875	0.4837	17.33	0.7542
		Standard LVGP	**0.5107**	**0.1410**	3.857	0.8379
		H-LVGP	0.5373	0.1523	**3.379**	**0.9048**
	80	Separate GPs	1.265	0.3418	7.985	**0.9243**
		Standard LVGP	0.3815	0.1041	2.251	0.8691
		H-LVGP	**0.3329**	**0.0969**	**2.001**	0.9888

Table 13: Branin — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.95). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

		Separate GPs				Standard LVGP				H-LVGP			
n	Level	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
25	$x_2 = 15$	5.902	0.1174	35.77	0.6200	6.136	0.1679	96.80	0.4100	**3.339**	**0.0917**	**32.68**	**1.000**
	$x_2 = 2$	14.17	0.4118	162.3	0.7950	**2.032**	**0.0488**	31.46	0.4300	2.327	0.0531	**19.96**	**1.000**
	$x_2 = 8$	7.824	0.8427	130.3	0.8600	3.853	0.3730	41.36	0.2900	**1.691**	**0.1164**	**30.41**	**1.000**
	$x_2 = 0$	16.19	0.2789	124.3	0.6850	**3.392**	0.0622	79.81	0.3450	3.565	**0.0615**	**28.74**	**1.000**
	$x_2 = 10$	12.14	0.6243	174.6	0.7100	4.247	0.2297	66.10	0.3800	**1.847**	**0.0949**	**32.76**	**1.000**
	Mean	11.24	0.4550	125.4	0.7340	3.932	0.1763	63.11	0.3710	**2.554**	**0.0835**	**28.91**	**1.000**
40	$x_2 = 15$	**0.5195**	**0.0128**	8.882	**1.000**	0.7171	0.0222	**6.774**	0.8400	1.016	0.0368	30.29	1.000
	$x_2 = 2$	6.850	0.2434	85.51	**0.9100**	3.729	0.1288	92.07	0.1650	**0.7563**	**0.0280**	**25.59**	1.000
	$x_2 = 8$	4.018	0.3915	38.87	**0.9050**	2.240	0.1564	**27.19**	0.4800	**0.8975**	**0.0521**	27.75	1.000
	$x_2 = 0$	3.333	0.0713	40.92	**0.9000**	2.888	0.0597	45.91	0.3400	**1.096**	**0.0212**	**34.81**	1.000
	$x_2 = 10$	6.753	0.2818	35.39	**0.9650**	3.958	0.1764	106.1	0.3000	**2.984**	**0.1347**	**34.58**	1.000
	Mean	4.295	0.2002	41.91	**0.9360**	2.706	0.1087	55.61	0.4250	**1.350**	**0.0546**	**30.60**	1.000

Table 14: Six-hump camel — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.95). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

		Separate GPs				**Standard LVGP**				**H-LVGP**			
n	Level	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
25	x_2 = -1	0.1689	0.2501	**2.355**	**0.9500**	**0.1532**	**0.2434**	3.163	0.5500	0.1713	0.2904	2.601	0.9250
	x_2 = -0.85	0.2249	0.3515	3.078	0.9600	**0.1706**	**0.2957**	4.223	0.6050	0.2511	0.2988	**1.745**	**0.9550**
	x_2 = 0	0.1822	0.4036	**1.563**	0.9800	**0.1337**	**0.3440**	2.544	0.5050	0.1679	0.4539	1.749	**0.9450**
	x_2 = 0.85	0.6129	0.6476	3.049	**0.9550**	**0.2916**	**0.3504**	7.908	0.1200	0.3352	0.4118	**2.640**	0.8900
	x_2 = 1	0.3711	0.3243	**2.151**	0.8250	**0.1729**	**0.2178**	3.481	0.3800	0.2401	0.3208	2.156	**0.9600**
	Mean	0.3120	0.3954	2.439	0.9340	**0.1844**	**0.2902**	4.264	0.4320	0.2331	0.3552	**2.178**	**0.9350**
40	x_2 = -1	0.1797	0.3163	2.472	**0.9350**	**0.0948**	**0.1151**	2.447	0.3600	0.1187	0.1803	**1.245**	1.000
	x_2 = -0.85	0.2187	0.3297	2.297	**0.9550**	0.2121	0.3042	5.321	0.2750	**0.1516**	**0.2230**	**1.854**	1.000
	x_2 = 0	0.1888	0.3379	2.033	0.8550	**0.0768**	**0.1458**	1.851	0.4000	0.0960	0.1751	**0.9064**	**1.000**
	x_2 = 0.85	0.1662	0.2421	**1.571**	**0.9950**	0.1337	0.1614	3.791	0.1900	**0.1159**	**0.1433**	1.920	1.000
	x_2 = 1	0.1719	0.2113	1.338	**0.9650**	**0.0972**	**0.1160**	2.485	0.3100	0.1162	0.1233	**1.278**	1.000
	Mean	0.1851	0.2875	1.942	**0.9410**	0.1229	**0.1685**	3.179	0.3070	**0.1197**	0.1690	**1.441**	1.000

Table 15: Griewank 6-D — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.95). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

		Separate GPs				Standard LVGP				H-LVGP			
n	Level	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
80	x6 = 0.5	**0.1124**	1.039	**1.094**	**0.8887**	0.1689	1.487	2.812	0.6250	0.1195	**0.9957**	1.528	0.6904
	x6 = 1.3	**0.0844**	**0.8664**	**0.8943**	**0.8369**	0.1409	1.420	2.162	0.6787	0.0939	0.9630	1.303	0.7451
	x6 = 2.7	**0.0423**	**0.9214**	0.5231	0.8672	0.0811	1.481	0.7551	0.8555	0.0542	1.033	**0.4656**	**0.9316**
	x6 = 3.1	0.0478	1.326	0.4536	0.8623	0.0852	2.148	0.6717	0.8643	**0.0461**	**1.280**	**0.3360**	**0.9678**
	Mean	**0.0717**	**1.038**	**0.7414**	**0.8638**	0.1190	1.634	1.600	0.7559	0.0784	1.068	0.9081	0.8337
120	x6 = 0.5	**0.1149**	1.039	1.405	**0.8096**	0.1157	**0.9518**	**1.175**	0.7100	0.1506	1.229	1.783	0.7100
	x6 = 1.3	**0.0656**	**0.7280**	**0.6875**	**0.9229**	0.0918	0.8605	0.8627	0.7793	0.1186	1.175	1.418	0.7754
	x6 = 2.7	**0.0354**	**0.7809**	**0.3830**	**0.9580**	0.0588	1.029	0.3864	0.9150	0.0709	1.299	0.6118	0.9688
	x6 = 3.1	**0.0355**	**1.005**	0.3884	0.8252	0.0636	1.727	**0.3583**	**0.9336**	0.0649	1.686	0.5136	0.9912
	Mean	**0.0628**	**0.8884**	0.7160	**0.8789**	0.0825	1.142	**0.6957**	0.8345	0.1012	1.347	1.082	0.8613
160	x6 = 0.5	**0.0731**	**0.7128**	**0.7801**	**0.9141**	0.1006	0.8151	0.8253	0.8291	0.1295	1.170	1.190	0.8584
	x6 = 1.3	**0.0658**	**0.7193**	0.8450	0.8428	0.0807	0.7678	**0.6693**	**0.9014**	0.1072	1.113	1.069	0.8711
	x6 = 2.7	**0.0496**	**0.9630**	**0.4770**	0.8779	0.0532	0.9725	0.4788	0.9082	0.0892	1.670	0.6728	**0.9600**
	x6 = 3.1	**0.0303**	**0.9437**	0.5085	0.7451	0.0457	1.228	**0.3392**	**0.9639**	0.0925	2.702	0.7174	0.9932
	Mean	**0.0547**	**0.8347**	0.6527	0.8450	0.0701	0.9459	**0.5782**	0.9006	0.1046	1.664	0.9121	**0.9207**

Table 16: Friedman 5-D (pilot) — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.95). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

		Separate GPs				**Standard LVGP**				**H-LVGP**			
n	Level	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95	MAE ↓	RRMSE ↓	IS ↓	Cov. → .95
40	level 1	2.883	0.7168	35.53	**0.5400**	**1.919**	**0.5056**	37.19	0.4258	1.928	0.5072	**34.30**	0.4678
	level 2	3.648	0.9445	25.21	**0.7588**	1.458	0.4125	19.22	0.5918	**1.420**	**0.4047**	**16.31**	0.6504
	level 3	2.100	0.5276	12.06	**0.9658**	1.061	0.2778	14.27	0.5918	**1.007**	**0.2648**	**9.959**	0.7246
	level 4	1.980	0.4667	23.02	0.5967	0.9341	0.2312	8.987	0.6963	**0.8655**	**0.2157**	**6.081**	**0.8389**
	Mean	2.653	0.6639	23.95	**0.7153**	1.343	0.3568	19.92	0.5764	**1.305**	**0.3481**	**16.66**	0.6704
60	level 1	2.159	0.5576	14.19	0.8428	**0.6205**	**0.1800**	5.883	0.7695	0.6499	0.1920	**4.671**	**0.8506**
	level 2	2.355	0.6243	23.10	0.6875	**0.5207**	**0.1553**	4.299	0.8213	0.5457	0.1697	**3.986**	**0.8486**
	level 3	1.977	0.4900	25.71	0.5654	**0.4212**	**0.1098**	**2.364**	0.8955	0.4548	0.1246	2.402	**0.9551**
	level 4	1.009	0.2630	6.315	0.9209	**0.4802**	**0.1189**	2.882	0.8652	0.4990	0.1227	**2.457**	**0.9648**
	Mean	1.875	0.4837	17.33	0.7542	**0.5107**	**0.1410**	3.857	0.8379	0.5373	0.1523	**3.379**	**0.9048**
80	level 1	1.456	0.4145	8.637	**0.9531**	0.4100	0.1139	2.381	0.8740	**0.3555**	**0.1131**	**2.024**	0.9795
	level 2	1.656	0.4538	10.23	0.8770	0.3519	**0.1006**	2.357	0.8340	**0.3254**	0.1023	**1.758**	**0.9756**
	level 3	1.157	0.3010	8.100	0.8789	0.3291	0.0890	**1.835**	**0.9023**	**0.2892**	**0.0786**	1.947	1.000
	level 4	0.7897	0.1980	4.970	**0.9883**	0.4351	0.1130	2.430	0.8662	**0.3615**	**0.0937**	**2.275**	1.000
	Mean	1.265	0.3418	7.985	**0.9243**	0.3815	0.1041	2.251	0.8691	**0.3329**	**0.0969**	**2.001**	0.9888